Agentic Pilot: An Evidence-Driven Local Autonomous AI Agent Framework for Privacy-Preserving Intelligent Task Automation

***Abstract*—Browser agents can report a task as finished when it is not, and many send every observation to a hosted model. Agentic Pilot is an open-source browser agent with local models that may report completion only when a rule-based predicate over the observed page holds, storing the inputs of each decision as evidence. We formalize this evidence-gated completion and its false completion and false rejection rates. *Measured at component level*, inside the agent's live verification node, the gate withheld 20 of 23 unsatisfied design states (95% interval 0.68–0.95) and rejected 1 of 15 satisfied ones; on 42 held-out states frozen before testing it withheld 9 of 22 (0.23–0.61) and rejected 1 of 20 (exact McNemar on paired correct decisions $p = 0.021$ against a dispatch-only rule), because its answer check depends on request wording. A decision costs a median of 27 ms and no model call. The model client refuses non-loopback endpoints, and in 70 episodes with a scripted stand-in for the model server the agent's Python processes connected only to loopback addresses. Memory retrieval has no relevance threshold, and recovery does not recognize a missing confirmation. *In a seeded simulation* built on these measurements and stated assumptions, the gate reduced episode-level false completions (19 against 28 of 100 synthetic scenarios) but did not significantly improve decision accuracy (52 against 46 of 74 claims, $p = 0.15$); these outcomes are conditional on the assumptions of a model corrected after an audit. *No end-to-end evaluation with a language model was executed*; its harness is released.**

***Index Terms*—LLM agents, web agents, local inference, privacy-preserving computing, completion verification, task automation, human-in-the-loop**

I. Introduction

Language-model agents now operate software for people: they open pages, fill in forms and read results [1]–[3]. Many send each observation, often a screenshot of a logged-in page, to a hosted multimodal model [2], [4], [5], although such screens carry names, messages and account data and personal information has been shown to leak from language models [6]. Independently of where the model runs, an agent can stop too early. We call this *premature completion*: a terminal "done" while the environment does not satisfy the goal, for example after opening the search engine but before issuing the query. Each action succeeded; the task did not.

a) Research gaps: We address three gaps, each stated so that it can be tested.

G1 *In-loop completion checks are rarely measured.* Benchmarks check the final state after the episode [1], [3], and judges assess saved trajectories [2], [7]; when the agent itself decides, it is usually its own model that does so [5], [8], and intrinsic self-correction is unreliable [9]. Whether a rule over observed state inside the loop withholds premature completions, and at what false-rejection cost, has not been reported with intervals and held-out data. Test: Sections VII-A and VIII-A.

G2 *The contribution of each kind of evidence is unknown.* If every conjunct of such a rule contributes, removing it should raise false completions. Test: the ablation of Section VIII-B.

G3 *Locality claims of local agents are rarely checked.* Running models locally is claimed to keep data on the machine, but whether an agent's process contacts other hosts is seldom measured. Test: the egress test of Section VIII-D.

b) Scope and terms: Agentic Pilot acts only in a Chromium browser. It is *autonomous* in that, given a request, it plans and executes browser actions without step-by-step instructions; the user approves tasks classified as high-risk (posting, sending, purchasing, deleting, transferring) and solves CAPTCHAs, and no end-to-end task with a language model is evaluated in this paper. It is *privacy-preserving* in a narrow sense: by default its model client sends prompts and screenshots only to a model server on the loopback interface, and it writes task records, memory and evidence only to local files (Section XI); whether that server runs inference locally depends on its configuration. Desktop, file and process control is a design (Appendix A).

c) Research questions: The gaps lead to five questions (Section VI): whether evidence-gated completion reduces false completions (RQ1, G1) and improves decision reliability (RQ2, G1, G2), whether memory helps repeated workflows (RQ3), whether evidence-guided recovery helps (RQ4) and what verification costs (RQ5).

d) Scientific contributions:

1) A formulation of evidence-gated completion that separates action dispatch from task completion, with false completion and false rejection rates (Section III).
2) ACTUAL evidence on G1 and G2: the gate's error rates on 92 constructed states in three sets, one frozen before testing, inside the live verification node, with intervals, paired tests and a per-conjunct ablation. Rule fixes made on design states transfer only partially (Section VIII).
3) A seeded, SIMULATION-based evaluation of H1–H5, with inputs measured on the implementation and every assumption stated, which indicates, under those assumptions, the trade-off between false completions and false

rejections and where the implementation falls short—memory without a relevance threshold, recovery blind to missing confirmations (Section IX).

e) Engineering contributions:

4) An open-source local agent with the gate inside its control loop (Algorithm 1, Section IV) and an implementation audit (Table II).
5) Enforcement of a loopback-only model endpoint and an ACTUAL test, with a stand-in model server, that the agent's Python processes connect only to loopback addresses, addressing G3 (Sections V and VIII-D).
6) An end-to-end harness: self-hosted sites with fictional content, 40 tasks with a dev/test split, an oracle validated on scripted trajectories, and baseline and ablation configurations (Section VII-B).

We do not claim that the agent completes tasks: the end-to-end evaluation is implemented but not run (Section VIII-E), and simulated results are labelled as such throughout.

II. Related Work

a) Reasoning-and-acting agents: ReAct interleaves reasoning with tool actions [8], building on chain-of-thought prompting [10]; Toolformer learns when to call an API [11]. Reflexion and RCI let the model critique and retry its own attempts [12], [13], and Voyager accumulates skills checked by a model [14]. In all of these a model decides that a task is finished, and intrinsic self-correction without external feedback does not reliably help [9].

b) Web and computer-use agents and benchmarks: Mind2Web pairs a ranker over Document Object Model (DOM) elements with an LLM [15]; WebVoyager drives live sites from screenshots [2]; SeeAct finds grounding to be the bottleneck [4], [16]. WebArena, VisualWebArena and WorkArena evaluate on self-hosted or remote-hosted applications with functional checks [1], [17], [18]; OSWorld and Windows Agent Arena do so in real operating systems [3], [19]. UFO, Agent S and AppAgent control desktop and mobile applications [5], [20], [21], and CogAgent, SeeClick, UI-TARS and OmniParser ground actions in screenshots [22]–[25]. These works say little about when the agent itself may conclude that a task is done.

c) Completion checking: Benchmark checkers run after the episode. WebVoyager evaluates trajectories with a multimodal judge [2], Pan et al. use model-based evaluators to judge and refine agents [7], and Sumyk and Kosovan train a completion judge for macOS applications [26]. We study a rule over observed state *inside* the loop, record its inputs, and measure its errors (G1, G2); a model judge is complementary.

d) Memory, security and local inference: Retrieval-augmented generation [27], [28], Generative Agents and MemGPT [29], [30] manage agent memory. Indirect prompt injection hides instructions in content the agent reads [31]; defenses are insufficient in systematic evaluations [32], InjecAgent, AgentDojo, ToolEmu and OS-Harm measure unsafe behavior [33]–[36], and CaMeL separates trusted control flow from untrusted data [37]. Open-weight models such as Qwen2.5 and Qwen2-VL run on consumer hardware [38]–[40]; surveys list trustworthiness among open problems of LLM agents [41], [42]. Table I compares representative systems.

TABLE I

Comparison with representative systems, from what each paper reports. Env.: W live websites, Wr self-hosted or cached pages, Ws simulated web, tool or text environment, D desktop OS, Mo mobile. Exec.: A agent, B benchmark, M model. Completion: who decides that a task is done—the agent's model, a benchmark checker after the episode, a model judge after the episode, or a rule over observed state inside the loop; n/a if the system does not end tasks. Evid.: persists per-step state artifacts. Mem.: memory across steps or episodes. Deploy.: L open-weight models used or supported, H hosted models only. Eval.: E2E end-to-end tasks, Off offline step prediction, C+S component tests and simulation. Y yes, P partial, N no, n.r. not reported. All listed systems released code.

System	Env.	Exec.	Completion	Evid.	Mem.	Deploy.	Eval.
ReAct [8]	Ws	A	model	N	N	H	E2E
Reflexion [12]	Ws	A	model[a]	N	Y	H	E2E
WebArena [1]	Wr	B	benchmark	n.r.	N	H	E2E
Mind2Web [15]	Wr	B	n/a	P	N	L	Off
WebVoyager [2]	W	A	model; judge[b]	Y	N	H	E2E
SeeAct [4]	W, Wr	A	model	n.r.	N	P	E2E, Off
OSWorld [3]	D	B	benchmark	n.r.	N	L	E2E
UFO [5]	D	A	model	n.r.	P	H	E2E
AppAgent [21]	Mo	A	model	n.r.	Y	H	E2E
CogAgent [22]	Wr, Mo	M	n/a	N	N	L	Off
MemGPT [30]	Ws	A	n/a	N	Y	H	E2E[e]
AgentDojo [34]	Ws	B	benchmark	n.r.	N	L	E2E
Agentic Pilot (ours)	Wr[c]	A	in-loop rule	Y	P[d]	L	C+S

[a] An evaluator gives feedback between trials. [b] Used for evaluation. [c] Evaluated only on constructed states and self-hosted sites; the implemented backend targets live websites, and a desktop backend is designed (Appendix A). [d] Implemented; studied only by simulation. [e] Document QA and conversation tasks, not web tasks.

III. Problem Formulation

a) Dispatch and completion: The agent observes an environment state s_t through an observation $\Omega(s_t)$ (for the browser: URL, title, element list and screenshot). A planner maps a goal g, a plan $P = (g_1, \ldots, g_n)$ and the observation to an action a_t. Executing it returns a *dispatch flag* $\alpha_t \in \{0, 1\}$, which says only that the call returned without error. The agent may report the task complete only when a completion predicate holds on the state observed *after* the action:

$$\Phi(g, s_{t+1}, P, \sigma) = \bigwedge_{c \in C(g,P)} c(s_{t+1}) \wedge \pi(P) \wedge \neg\beta(s_{t+1}, \sigma), \quad (1)$$

where $C(g, P)$ is the set of postconditions that the goal and plan imply, $\pi(P)$ states that no plan step is pending, and β is a blocking condition (a bot-check page, an unapproved high-risk action) with σ the approval state. Dispatch does not imply completion, $\alpha_t = 1 \not\Rightarrow \Phi = 1$. Every evaluation of Φ is stored with the observed values it used. The same form applies to desktop, file and process state, whose postconditions are typed and exact (Appendix A); only the browser instance is implemented.

b) The implemented browser predicate: $C(g, P)$ is derived from the wording of the request. Φ is the conjunction of:

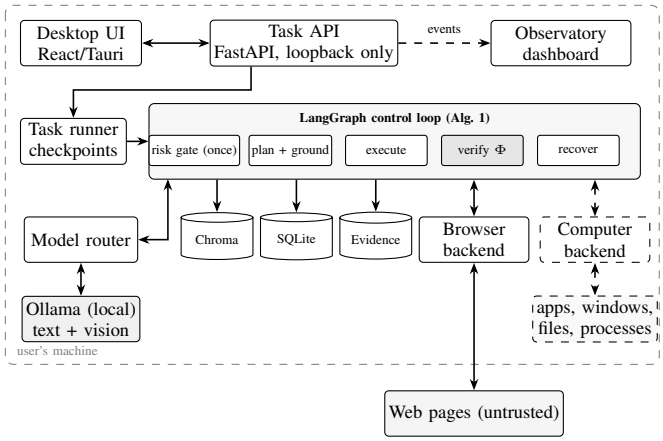


Fig. 1. Components of Agentic Pilot. Solid: implemented; dashed: proposed. Everything inside the boundary runs on the user's machine; the model endpoint must be a loopback address unless the user overrides it. The browser backend is the only implemented execution environment; the computer backend and the local resources it would act on are a design (Appendix A). The Observatory also reads the task database and evidence files.

no browser-error URL, error or "not found" title, or redirect to a sign-in page the request did not name (E); not still on a search engine's home page when a search, an extraction or a multi-step plan was requested (H); a results page carrying the query quoted in g (Q); the literal text that g asks to be entered present in an input, exactly if g says "exactly" (T); for extraction requests, an answer or structured extraction that mentions a content word of g (X); and, for multi-step plans, no pending step unless a structured extraction exists (R). Each conjunct is instantiated only when its trigger words occur, so for a request that triggers none, Φ reduces to $\neg E$. A pre-check implements β for bot-check pages and halts the task as BLOCKED. A plan step counts as done when an action other than `complete` dispatches during it. Φ has no tunable threshold: its operating point is fixed by the rules, so we characterize the trade-off between false completions and false rejections by removing conjuncts (Section VIII-B) rather than by a curve.

c) Metrics: Let $\mathcal{T}$ be a set of episodes or end states, $\mathcal{T}_c \subseteq \mathcal{T}$ those reported complete, and $\Phi^\star$ an oracle written independently of Φ (Section VII). With $N_{\mathrm{FC}} = |\{\tau \in \mathcal{T}_c : \Phi^\star(\tau) = 0\}|$ and $N_{\mathrm{FR}} = |\{\tau \in \mathcal{T} \setminus \mathcal{T}_c : \Phi^\star(\tau) = 1\}|$,

$$\mathrm{FCR} = \frac{N_{\mathrm{FC}}}{|\mathcal{T}_c|}, \quad \mathrm{FRR} = \frac{N_{\mathrm{FR}}}{|\{\tau \in \mathcal{T} : \Phi^\star(\tau) = 1\}|}. \quad (2)$$

Because a dispatch-only rule reports every state complete, we also report the share of unsatisfied states that the gate withholds, $1 - N_{\mathrm{FC}}/|\{\tau : \Phi^\star(\tau) = 0\}|$. FCR alone rewards an agent that never finishes, so the rates are always reported together, as counts with Wilson 95% intervals [43].

IV. ARCHITECTURE

Fig. 1 shows the components. A desktop client submits a natural-language task to a FastAPI service bound to the loopback interface. The task runner executes the control loop, checkpoints its state after every node and cancels tasks that exceed a wall-clock limit (5 min by default). Task events go to SQLite and to a read-only monitoring process.

Algorithm 1 Evidence-gated control loop (browser backend)

Require: goal g; budget of K=15 model calls
$I \leftarrow$ PARSEINTENT(g); $P \leftarrow$ DECOMPOSE(g) if multi-step
if I is high-risk and not approved **then return** WAITAPPROVAL
end if
NAVIGATE($I.site$); on failure go to line 14
loop
  $o \leftarrow$ OBSERVE ▷ sanitized DOM, screenshot
  if $\beta(o)$ **then return** BLOCKED ▷ hand off to user
  end if
  $a \leftarrow$ PLAN($g, P, o,$ RETRIEVE(g)) ▷ model call
  $\alpha \leftarrow$ EXECUTE(a); save before/after evidence
  if $\alpha = 0$ **then**
    if fewer than 3 retries **then continue**
    end if
    st $\leftarrow$ RECOVER(error)
    if *st* is exhausted or blocked **then return** FAILED
    end if
    continue ▷ retry, vision, ...
  end if
  if $a \neq$ `complete` **then** mark step done; **if** steps remain **continue**
  end if
  if $a =$ `complete` or last step or text typed **then**
    $(v, \mathrm{obs}) \leftarrow \Phi(g, s, P)$; save as evidence
    if v **then** store memory; **return** COMPLETED
    end if
  end if
  if model calls $\geq K$ **then return** FAILED
  end if
end loop

A. Control Loop

The loop is a compiled LangGraph state machine with eleven nodes; Algorithm 1 gives its behavior. A typed state record carries the intent, the plan and step index, the observation, the planned action, the action history, retry and model-call counters, routing decisions and the status. Only the verification step can end a task as COMPLETED, and only when Φ holds (Fig. 2).

a) Planning: A small model parses the request into an intent (action, site, target, content, risk level). Requests with a connective or more than eight words are decomposed into steps with an action type and expected outcome. Each iteration, the planner sees the goal, the plan, the current observation and retrieved memory and returns one action. Deterministic rules then adjust it: exact-text requests are grounded to the best visible text field, repeated navigation becomes a failure, and plan steps that name clicking or extracting map to those actions.

b) Recovery, routing and memory: A failed action is retried through a fresh observation up to three times; later failures are classified into seven types and assigned a strategy (retry, alternative selector, vision fallback, replan), with at most six attempts. Only the vision-fallback strategy changes behavior, by routing the next step to the vision model; the other strategies repeat ordinary planning. A deterministic router assigns each model call a role and the first installed candidate among local Qwen2.5, Qwen3.5 and DeepSeek-R1 models

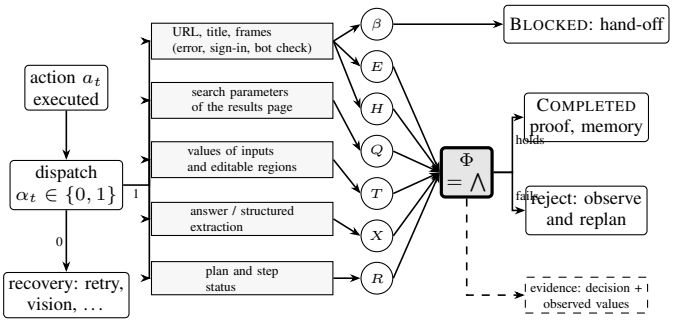


Fig. 2. Evidence-gated verification after an action (Algorithm 1, lines 7 and 22). A failed dispatch goes to recovery and never to Φ. A successful dispatch makes the observed values available to the conjuncts of the browser predicate, each triggered by the wording of the goal (Section III); the bot-check condition β is tested first and hands the task to the user. Every decision is stored with the values it used.

(1.5B–7B parameters), or `qwen3-vl:2b` and `moondream` for vision. At termination, outcomes are written to a local Chroma collection and retrieved into later planning prompts.

B. Browser Backend

A pool of Playwright contexts over one Chromium instance gives each task its own page. Each observation is a DOM pass over interactable elements, recording tag, role, label, up to 120 characters of text, type, visibility, bounding box and selectors; elements are ranked, deduplicated and capped at 50, and their text is passed through a prompt-injection sanitizer. Actions are navigate, click, type, key press, scroll and extract; typed text is read back. The vision model is used only when the planner returns `need_help` or recovery requests it: it receives a screenshot and returns a point, which is clicked without an individual check. Per task, the evidence directory holds before/after screenshots of each action, the latest DOM manifest, the verification and recovery records and the node trace.

V. IMPLEMENTATION

The backend is Python (about 11,000 lines) on FastAPI, LangGraph, Playwright and Pydantic, with SQLite and a persistent local Chroma store; model calls use the Ollama HTTP API with JSON output and a fixable temperature and seed. The client is React in a Tauri shell. The code is public under the MIT license at https://github.com/sudeshsudhii/Agentic-pilot-local; this paper describes branch `ccr-4b8200f0-5i3k5m`, whose agent code (`backend/`) is unchanged since commit `bf0a8a5`; the experiment scripts, data and simulation are on the same branch. The repository's default branch has diverged (it defaults to a hosted model and adds desktop automation) and is not the version evaluated here; a release tag for the evaluated version has not yet been created.

a) Component audit: Agent documentation tends to run ahead of the code, so we traced each component to the code path that executes it and to the tests and experiments that exercise it (Table II). Claims in this paper are restricted to rows with evidence in the last column; the other rows are implemented but unevaluated, or designs.

TABLE II
IMPLEMENTATION AUDIT, TRACED THROUGH THE CODE. CLASS: IE IMPLEMENTED AND EXPERIMENTALLY EVALUATED (COMPONENT LEVEL, ACTUAL); IS IMPLEMENTED, STUDIED ONLY BY SIMULATION; IN IMPLEMENTED, NOT EVALUATED; PI PARTIALLY IMPLEMENTED; PR PROPOSED. EVIDENCE: C COMPONENT STUDY, O ORACLE VALIDATION, E EGRESS TEST WITH A STAND-IN MODEL SERVER, T UNIT TEST, M MEASURED INPUT TO THE SIMULATION, S SIMULATION.

Component	Class	Evidence
Completion predicate Φ, bot-check halt	IE	C (92 states), M, S
Browser control (Playwright), DOM extraction	IE	C, M (latency)
Loopback-only model endpoint; Chroma telemetry off	IE	E (70 stand-in episodes), T
End-to-end harness and oracle $\Phi^\star$	IE	O (114 trajectories)
Recovery engine: classification, strategy order	IS	M (6 failure modes), S
Memory store retrieval (Chroma, SQLite)	IS	M (10 queries), S
Per-action evidence (screenshots, records)	IS	M (latency), S
LLM planner in the full loop	IN	tests with mocked model
Vision fallback; task-level human approval	IN	tests
Recovery: alternative selector, replan	PI	labels only
Self-verification baseline mode	IN	harness only
Enhanced validation; evidence-guided recovery; memory relevance filter	PR	S (assumed behaviour)
Desktop backend, action gate, permission tiers	PR	–

b) Fixes: The audit and the gate study led to fixes. At commit `2361d50`, the extraction handler formatted every answer as facts about one university campus and substituted fixed text when a page had too few matching lines; it now keeps only page lines that share terms with the goal. The verification node accepted the reasoning of any action as an answer and let a `complete` action mark a pending step done; both are fixed, and the Q, sign-in and "not found" checks, exact matching and goal-term overlap were added. Bot-check detection no longer fires on titles that contain "robot" or on hidden frames. The vision-fallback strategy, memory retrieval and the verification switch were inert and now take effect; routing keywords match whole words. The repository's earlier "certification" figures counted a task as passed when it ended as completed and a screenshot existed, and we use none of them.

c) Locality enforcement: The model client now refuses any endpoint whose host is not a loopback name or address, before connecting, unless the user sets an explicit override; LAN addresses count as remote. Chroma stores are created with telemetry disabled, and the privacy audit log classifies destinations by parsed host instead of by substring. For experiments we added switches that replace Φ by no check or by a self-check of the planner model, remove single conjuncts, or pin one recovery strategy, and the gateway records the token counts that Ollama reports.

VI. RESEARCH QUESTIONS AND HYPOTHESES

The gaps of Section I lead to five research questions: does evidence-gated completion reduce false completions (RQ1, G1) and make completion decisions more reliable (RQ2, G1–G2); does memory improve consistency on repeated workflows (RQ3); does evidence-guided recovery improve recovery

TABLE III
HYPOTHESES WITH NULL (H_0) AND ALTERNATIVE (H_1) FORMS. CONTROLS FOR ALL: SAME SCENARIOS, PLANNER BEHAVIOUR, BUDGET AND SEEDS. TEST: EXACT MCNEMAR ON PAIRED SCENARIO OUTCOMES, WILSON INTERVALS, MONTE CARLO OVER REGENERATED SETS. LAST COLUMN: OUTCOME IN THE SIMULATION (SECTION IX); IT IS NOT AN EMPIRICAL RESULT.

	H_1 (H_0: no difference)	IV	DV	Simulated outcome
H1	Evidence-gated verification lowers false completions vs. accepting every claim	Φ on / off	FC per scenario, FCR	lower by construction; size 19 vs 28 FC (cond.)
H2	Evidence validation raises decision reliability	Φ on / off; enhanced	decision accuracy	not supported for Φ (accuracy .70 vs .62, p=.15 cond.); enhanced: yes by assumption
H3	Memory raises contextual consistency on repeated workflows	memory off / on / filtered	TSR, ctx. accuracy	not supported; underpowered (19 vs 19 of 44)
H4	Evidence-guided recovery recovers more failures than retry	recovery mode	recovered share	higher by assumption (87 vs 67 of 93; proposed)
H5	Verification adds overhead and makes decisions more conservative	verifier	time, FRR	small time (+2.6% mean), FRR higher

(RQ4); and what does verification cost (RQ5)? Table III states the corresponding hypotheses. At the level of the full agent each is tested here only by simulation (Section VII-C), which shows what the hypotheses imply under stated assumptions; H1 and H2 also have ACTUAL component-level evidence from the gate study (Section VIII-A). *None is confirmed empirically end to end*; the protocol of Section VII-B is that test. For H4 and the enhanced variant of H2 the simulated outcome follows directly from the assumed behaviour of unimplemented components, so their synthetic p-values carry no information beyond those assumptions.

VII. EVALUATION METHODOLOGY

The experiments need no language model. The current-code results were produced on 2026-10-04 on a Linux container with a 4-core Intel Xeon at 2.1 GHz, 15 GiB RAM and no GPU, with Python 3.11.15, Playwright 1.56.0, headless Chromium 141 and LangGraph 1.2.12; the decisions of the code before the fixes come from an earlier run on a similar container and are deterministic. Scripts, seeds and raw outputs are under `artifacts/` and `benchmarks/e2e/` in the repository.

A. Gate Study

a) States and splits: A state is a goal, a URL, an HTML document served through Playwright request interception, the agent's answer and plan state where relevant, and an oracle label. Every state ends with a dispatched action, so a dispatch-only rule accepts all of them. The *design* set (38 states in six categories) was used to find the gate's defects and design the fixes. *Held-out 1* (12 states) was written after the fixes and not used to change them. *Held-out 2* (42 states, 22 unsatisfied) was written to a fixed procedure—varied user phrasings per category, including wordings the gate's trigger lists do not contain, with a balanced split—and committed before the gate was run on it; afterwards only configuration switches for the ablation were added to the gate code, which leave its default behaviour unchanged. Its SHA-256 is recorded in the repository.

b) Oracle: The label $\Phi^\star$ of a state states whether the goal is achieved by the page and answer shown, judged from the page's facts (for example, whether the answer contains the year in the article). It was written together with the state, before the state was judged by the gate, and it is computed by no code shared with Φ. Its limitation is that one author wrote all states and labels, so no inter-annotator agreement exists; a protocol for independent relabelling with Cohen's κ [44] is in the repository.

c) Modes and statistics: Each state is judged by the pre-check and Φ called directly, and by the real verification node (database, browser-pool and evidence I/O stubbed) with an empty or a generic one-sentence planner reasoning. For the ablation, each conjunct of Φ and the pre-check β is removed in turn by a configuration switch. Counts carry Wilson and Clopper–Pearson 95% intervals [43], [45]; comparisons on the same states use the exact McNemar test [46] on paired decision correctness. Against the dispatch-only rule the discordant pairs are the withheld unsatisfied states (only the gate is correct) and the rejected satisfied states (only dispatch is correct); an earlier version tested the withheld count alone and is corrected here. Latency is the median of 20 repetitions after a warm-up.

B. End-to-End Harness

a) Environment and tasks: Live websites change and block automated traffic, and WebArena-style site images were not available offline, so we built five self-hosted sites served on the loopback interface with fictional content that a model cannot answer from memory: a search engine, an encyclopedia, a shop with search and cart, a town-office contact form and newsletter, and a notes application; a further path redirects to a sign-in wall. The 40 tasks (10 development, 30 test) cover search, navigation, text entry, extraction, form submission, multi-step tasks and 5 unsatisfiable requests, at four difficulty tiers by the number of actions a solution needs (1, 2–3, 4–6, more). The notes site appears only in the test split. Tasks needing an account, payment, a CAPTCHA, the live web, or an outcome that neither server state nor answer text can verify were excluded by design.

b) Oracle: $\Phi^\star$ is computed from signals that neither the agent nor Φ uses: server-side records of the sites (cart contents, form submissions, saved notes, field values autosaved by page scripts), the final URL, and the ground-truth facts in the site data matched against the agent's answer; unsatisfiable tasks are never achieved. It calls no model and shares no code with the verifier. We measured its error rate on 114 scripted trajectories driven directly through Playwright without the agent: a correct solution for each task, an idle trajectory that only opens the start page, and a near miss for 34 tasks (wrong value, wrong item, extra submission, partial query).

c) Protocol: Table XV (Appendix B) lists the configurations: the full system, two baselines—(a) no completion check and (b) self-verification by the planner model—and one-factor ablations of each conjunct, memory, recovery and each recovery strategy. Each configuration runs in its own process with its own database and memory store, at temperature 0 with a fixed seed, three trials per task, and the same budget of 15 planner calls. Per episode the harness records the oracle verdict, the agent's status, browser actions, model calls and tokens, wall-clock time, each verification decision with its latency, recovery strategies, retrieved memories, Ollama version and model digests, and every host the browser contacted. The analysis reports success, FCR and FRR with Wilson intervals, per-category breakdowns, exact McNemar tests and task-level bootstrap intervals [47] for differences to the full system, and the overhead of Φ relative to baseline (a). A published agent framework as a third baseline would need its own model and prompts and is left to the authors' runs.

C. Simulation-Based Evaluation and Data Provenance

a) Why simulation: The end-to-end protocol needs a local language model, which could not be installed in the environment used for this paper. To study the questions of Section VI before those runs exist, we built a seeded simulation of the agent's completion decision, recovery, memory and latency (`benchmarks/simulation`). The simulation does not execute the agent. Its outputs follow from two kinds of input, and every number in this paper carries one of five classes: ACTUAL (measured on the implementation), DERIVED (computed from actual measurements), SYNTHETIC (produced by the simulation), LITERATURE, or PROPOSED (behaviour of an unimplemented component). Tables and figures with synthetic values say so in their captions, and an evidence ledger lists the simulated values reported in the main tables with their source and class.

b) Measured inputs: Without a model we measured, on the implementation (2026-10-23, 4-core Intel Xeon at 2.8 GHz, same software versions): the decisions of Φ and of a basic success detector (error check E only) on the 92 gate states; the latency of navigation, clicks, form filling, DOM extraction, screenshots and Φ on the self-hosted sites (30 repetitions each); the failure class and strategy sequence that the real recovery engine assigns to six failure messages; and the ranks at which the real Chroma memory store returns relevant records for ten queries over a 22-record synthetic memory corpus. Detection rates used by the simulation are derived from the two held-out gate sets only (the 22 and 5 unsatisfied states, without the bot-check category, for detection; the 24 satisfied states of the same categories for false rejection), with the estimator $(k+0.5)/(n+1)$, which gives a small nonzero rate (0.07–0.13) where none was observed, as for basic detection in five categories and Φ on extraction. Simulated categories map to gate categories as follows: navigation, search, extraction and multi-step to themselves, data entry to text entry, and form submission to text entry and multi-step pooled. The detection rate of a missing confirmation is proxied by the multi-step rate, which rests on 3 states.

c) Assumptions and generation: What needs a model—how often the planner claims completion of an unfinished task, how often actions fail and are recovered, how much memory helps—is assumed and listed with its class in Table XVI (Appendix C). A scenario has a task category (navigation, search, form submission, data entry, extraction, multi-step) and an outcome class (successful, incomplete, incorrect action, missing evidence, contradictory evidence, stale evidence, ambiguous) that sets the probability that the goal is actually met. Each scenario draws its random numbers once, and every configuration reuses them, so comparisons are paired by scenario. The true outcome is evaluated per configuration because memory and continuation after a rejection change the final state. Parameters and code were committed (`3a0aa8e`) before the reported held-out set (seed 202, 100 scenarios) was generated; a development set (seed 101) was used only to debug the code. After the freeze, one function was changed: the latency model (Study F) did not give all configurations the same random draws, so it was corrected to common random numbers in `41a6097`; Studies A–E and H were not affected and gave identical outputs before and after. Later, `episode()` gained three output fields (claim, true state at the claim, first decision) for the decision-level analysis; outputs were otherwise byte-identical.

d) Corrected model (v2, post hoc): A methodology audit of the frozen model (v1) found that it contradicted the implementation in four places: its false-rejection rates came from bot-check pages, a category the scenarios do not contain; a rejected claim followed by a successful continuation was accepted without a second check, although the verification node checks every claim; the basic detector did not fail closed on an unreadable page, although it shares the error check E; and recovery was applied to every action failure, although action errors are first retried three times inline and a missing confirmation never reaches the recovery engine. The results reported here come from a corrected model, v2 (`simulate_v2.py`), that fixes these four points, draws derived rates from their Beta posteriors in the Monte Carlo replications, adds re-planning to the latency model and sweeps the continuation rate. v2 reuses v1's generator, seeds, scenario set and assumed values. It was written after the v1 results were known and is therefore *not pre-registered*; v1's outputs are archived (`out/heldout_v1`) and every number that changed is listed in the statistical correction report. Monte Carlo replications regenerate the scenario set 1,000 times to show the variability the generator and the rate uncertainty induce.

e) Metrics: With TP, FP, FN, TN the confusion counts of reported completion against the true outcome over N scenarios: $\mathrm{TSR} = \mathrm{TP}/N$; FCR and FRR as in Eq. (2), so that precision $= 1 - \mathrm{FCR}$ and recall $= 1 - \mathrm{FRR}$; F1 and accuracy $(\mathrm{TP}+\mathrm{TN})/N$. Recovery success is recovered over recoverable failures; retrieval quality is precision and recall at $k = 3$ and the mean reciprocal rank (MRR) of the first relevant

record. Episode time is decomposed as

$$T = T_{\text{inf}} + T_{\text{browser}} + T_{\text{evidence}} + T_{\text{verify}} + T_{\text{recovery}}, \quad (3)$$

with browser, evidence and Φ times resampled from the measurements and inference time assumed. Two levels are kept apart. *Policy-level* outcomes describe how an episode ends (task success, episode-level false completion); *decision-level* outcomes describe whether the verifier's first decision on a completion claim matches the true state at that claim. Because a rejection can change how an episode ends, the final ground truth of a scenario may differ between configurations, so only the decision level compares verifiers against one shared truth. Paired configurations are compared by the exact McNemar test on per-scenario binary indicators of one level, and the variability of paired differences across regenerated scenario sets by Monte Carlo percentiles. Every p-value and interval computed from synthetic data is conditional on the simulation assumptions; it describes the synthetic sample, not the deployed agent.

f) Studies: Study A generates the scenarios and computes the completion metrics, B compares four completion rules, C ablates components, D evaluates memory, E recovery, F overhead, G privacy (analytically, Section XI) and H sensitivity.

g) Independent validation: For v1, a separate validation agent recomputed, with independently written code, the confusion counts, rates and paired tests of Studies B and C and the memory and recovery statistics from the per-scenario files, and found no discrepancy (`simulation_validation_report.md`). Monte Carlo, sensitivity, latency, decision-level and recovery paired-test values were not independently recomputed. This check is a recomputation of the analysis code's arithmetic from the same synthetic files; it is neither independent human annotation of any label nor an independent replication of an experiment.

VIII. Results

All results concern the browser backend and its components. None measures end-to-end task success.

A. Completion Gate (G1)

Table IV reports the gate study. On the code before the fixes, the pre-check and Φ withheld 15 of the 23 unsatisfied design states and rejected 3 of the 15 satisfied ones; inside the live verification node they withheld only 14 (empty planner reasoning) or 13 (verbose reasoning), because the node let a premature `complete` mark a pending step done and took the planner's reasoning for an answer. After the fixes all three modes agree on every state of every set. On the design states the gate withholds 20 of 23 and rejects 1 of 15, and corrects 7 states relative to the earlier gate without breaking any; we report these counts without significance tests, because the fixes were designed on these states and a test of them would be circular.

The held-out sets give the honest estimate. On held-out 2, frozen before testing, the gate withholds 9 of 22 unsatisfied states (0.41; 95% interval 0.23–0.61) and rejects 1 of 20 satisfied ones (0.05; 0.01–0.24), so it is correct alone on 9 states and wrong alone on 1 (exact McNemar $p = 0.021$); pooled with held-out 1, it withholds 11 of 27 (0.25–0.59) and rejects 2 of 27 (0.02–0.23; 11 against 2, $p = 0.022$). On held-out 1 alone (2 against 1) the test has no power ($p = 1.0$). Its misses follow from the rules. The answer check X is triggered by words such as "extract" and "information about", so requests phrased "tell me", "what year", "find" or "get me" are not checked at all, and when X fires it accepts any answer that repeats a goal term ("Tokyo is the capital of Japan" for the population of Tokyo): none of the 5 unsatisfied extraction states is withheld. Text-entry requests phrased "write … in" or "fill … with" are not parsed; error pages titled "410 Gone" or "Something went wrong", a redirect to a different page, a press-and-hold human check, a results page for another query requested as "Google '…'", and an add-to-cart click that added nothing all pass. The one false rejection is a page that merely mentions reCAPTCHA. Hand-written rules thus fix the cases they were written for and generalize only partially, which answers G1 with a qualified yes: the gate withholds a significant share of premature completions, with a false-rejection rate whose 95% upper bound is 0.24, but far fewer than on the states it was designed on.

TABLE IV

Completion gate on constructed end states. n^-/n^+: states that do not / do meet the goal under the oracle. W: unsatisfied states withheld (a dispatch-only rule withholds none); FR: satisfied states rejected; FC/R: false completions over states reported complete (FCR); it depends on the constructed share of unsatisfied states and is not an estimate for real use, unlike W/n^- and FR/n^+. Brackets: Wilson 95% intervals. p: exact McNemar test on paired decision correctness against the dispatch-only rule on the same states (discordant pairs: W states only the gate decides correctly, FR states only the dispatch-only rule does); it is not computed on the design set (–), because the rules were written on those states. Upper block: current gate by category; lower block: totals, including the gate before the fixes. Held-out 2 was frozen in a commit before the gate was run on it. Source: `gate_stats.csv` (column `p_correct_vs_dispatch`).

	Design (38 states)			Held-out 2 (42 states)		
Category	n^-/n^+	W	FR	n^-/n^+	W	FR
Search	4/3	4	0	3/3	2	0
Navigation	6/2	5	0	4/4	1	0
Bot check	4/2	3	0	4/2	3	1
Text entry	3/3	3	0	3/3	1	0
Extraction	4/3	4	1	5/5	0	0
Multi-step	2/2	1	0	3/3	2	0

Set	Gate	W / n^-	FR / n^+	FC/R	p
Design	before	15/23 [.45,.81]	3/15 [.07,.45]	8/20	–
Design	after	20/23 [.68,.95]	1/15 [.01,.30]	3/17	–
Held-out 1	before	1/5 [.04,.62]	1/7 [.03,.51]	4/10	1.0
Held-out 1	after	2/5 [.12,.77]	1/7 [.03,.51]	3/9	1.0
Held-out 2	after	9/22 [.23,.61]	1/20 [.01,.24]	13/32	.021
Held-out 1+2	after	11/27 [.25,.59]	2/27 [.02,.23]	16/41	.022

TABLE V
ABLATION OF THE GATE ON ALL 92 CONSTRUCTED STATES: FALSE COMPLETIONS (FC) AND FALSE REJECTIONS (FR) WITH ONE CONJUNCT OF Φ, OR THE BOT-CHECK HALT β, REMOVED. p: EXACT MCNEMAR TEST OF CORRECT DECISIONS AGAINST THE FULL GATE ON THE DESIGN STATES (PAIRED BY STATE). ON THE HELD-OUT SETS NO REMOVAL CHANGES MORE THAN THREE DECISIONS. SOURCE: VERIFICATION_GATE_ABLATION.CSV.

	Design (23/15)			Held-out 1 (5/7)		Held-out 2 (22/20)	
Removed	FC	FR	p	FC	FR	FC	FR
none (full Φ)	3	1	–	3	1	13	1
E error, sign-in	8	1	.06	4	1	14	1
H search home	7	1	.12	3	1	15	1
Q query match	4	1	1.0	3	1	13	1
T typed text	6	1	.25	3	1	14	1
X answer	6	0	.62	3	1	13	1
R pending steps	4	1	1.0	3	1	15	1
β bot check	5	1	.50	4	0	15	0

B. Contribution of Each Conjunct (G2)

Table V removes one conjunct at a time. On the design states every removal raises false completions, most for the error check E (3→8) and the search-home check H (3→7), but with 23 unsatisfied states no single removal reaches $p < 0.05$ ($p \geq 0.06$), which with so few states is not evidence that a conjunct is unnecessary. On the held-out sets each removal changes at most three decisions, and removing Q or X changes none: these conjuncts rarely fire on phrasings they were not written for. Only two removals trade false completions for false rejections: the bot-check halt β (held-out 2: 13→15 FC, 1→0 FR) and, on the design states, the answer check X (3→6 FC, 1→0 FR). Evidence types thus contribute on the states that trigger them, and the gate's coverage, not the strength of any single check, limits it.

C. Overhead

Φ makes no model call and no browser action; it reads the page through the browser protocol. Over the 83 decisions that evaluate Φ the median latency was 26.8 ms (90th percentile 29.5 ms, 95th 30.3 ms, maximum 34.6 ms; percentiles of per-state medians of 20 repetitions, which understate the tail of single decisions), and decisions stopped by the pre-check took at most 13.4 ms. Its added tokens and API cost are zero; the added steps that a rejection causes can only be measured end to end (Section VIII-E).

D. Locality (G3) and Oracle Validity

To test locality we ran 70 agent episodes (the 10 development tasks under 7 configurations, six with memory enabled) through the real control loop and Chromium against the local sites, with a scripted stand-in for the Ollama API on 127.0.0.1:11435, on an uncommitted working tree based on commit `1435fe9` (recorded in the run's metadata). We logged every outbound `connect` call made from the harness's Python processes; the Playwright driver, Chromium, the model server and DNS lookups were not instrumented. All 555 attempts went to 127.0.0.1, all to the stand-in's port, and every page request that Playwright reported named host 127.0.0.1; WebSockets and browser-internal traffic were not observed. Because the stand-in claims completion at once, the episodes made no browser action beyond the first navigation, never called the vision model and never retrieved documents; memory retrieval ran in 54 of them. The stand-in always claims completion, so these episodes say nothing about task success. The repository's test suite runs a reduced version of this check (three development tasks, one configuration, memory and retrieval disabled), together with tests that a remote model address is refused and that Chroma telemetry is off; 144 of its 147 tests pass; the 3 failures need live websites or a running local model.

The end-to-end oracle agreed with the intended verdict on all 114 scripted trajectories: no false negative among 37 achieved trajectories (95% interval 0–0.09) and no false positive among 77 unachieved ones (0–0.05). The trajectories were written by the oracle's author, so this bounds the oracle's errors on anticipated behaviors only.

E. End-to-End Evaluation

We have not run the end-to-end evaluation with a language model: the environment used for this paper could not download model weights. Consequently this paper makes no claim about task success, about the gate's effect on success or on false completions of the full agent, about the baselines, or about the contributions of memory, retrieval and the recovery strategies; Table II marks these components as unevaluated. The harness, task set and analysis that produce these measurements are released and run with one command on a machine with Ollama.

IX. SIMULATION-BASED EVALUATION

All values in this section except those marked ACTUAL are SYNTHETIC: they come from the corrected seeded simulation (v2, post hoc) of Section VII-C, not from running the agent. They show the consequences of the stated assumptions and the measured component behaviour; they do not establish how the deployed agent performs. Every p-value and interval here is conditional on those assumptions.

A. Baselines and Verification (Studies A and B; H1, H2)

Table VI compares four completion rules on the same 100 held-out scenarios at two levels that must not be mixed. At the *decision level*, the planner claims completion in 74 scenarios, 46 of them truly complete at that moment; the four verifiers face the same 74 claims and the same truth, so their decisions are paired. At the *policy level*, an episode's final state may differ between configurations for the same scenario, because a rejected claim returns control to planning and the episode may then finish (or, after a false rejection, never be reported); task success and episode-level false completions are paired outcomes of different policies, not of one ground truth.

H1 (false completions). Episodes end in a false completion 19 times with Φ against 28 without verification (FCR 0.30 against 0.38; discordant pairs 9 and 0, exact McNemar $p = 0.004$; Table VII); basic success detection ends in 24.

TABLE VI

SIMULATION-BASED (SYNTHETIC) STUDY B ON THE SAME 100 HELD-OUT SYNTHETIC SCENARIOS (CORRECTED MODEL V2). POLICY LEVEL: FINAL OUTCOME OF EACH EPISODE (TP REPORTED AND ACHIEVED, FP REPORTED BUT NOT ACHIEVED, FN ACHIEVED BUT NOT REPORTED, TN NEITHER); THE FINAL STATE CAN DIFFER BETWEEN CONFIGURATIONS BECAUSE A REJECTION RETURNS CONTROL TO PLANNING. DECISION LEVEL: THE VERIFIER'S FIRST ACCEPT/WITHHOLD DECISION ON THE 74 SCENARIOS IN WHICH THE PLANNER CLAIMS COMPLETION (46 TRULY COMPLETE), AGAINST THE TRUE STATE AT THAT CLAIM. WILSON 95% INTERVALS. SYNTHETIC p-VALUES AND INTERVALS ARE CONDITIONAL ON THE SIMULATION ASSUMPTIONS; THEY ARE NOT EMPIRICAL TESTS OF THE AGENT. ‡NOT IMPLEMENTED (PROPOSED).

Configuration	TP	FP	FN	TN	TSR	FCR	FRR	Decision TP/FP/FN/TN	Decision acc.
No verification	46	28	3	23	46/100 [0.37,0.56]	28/74 [0.28,0.49]	3/49 [0.02,0.17]	46/28/0/0	46/74 [0.51,0.72]
Basic success detection	43	24	6	27	43/100 [0.34,0.53]	24/67 [0.25,0.48]	6/49 [0.06,0.24]	43/24/3/4	47/74 [0.52,0.74]
Evidence-gated (Φ)	44	19	6	31	44/100 [0.35,0.54]	19/63 [0.20,0.42]	6/50 [0.06,0.24]	43/19/3/9	52/74 [0.59,0.79]
Enhanced validation‡	46	8	8	38	46/100 [0.37,0.56]	8/54 [0.08,0.27]	8/54 [0.08,0.27]	41/8/5/20	61/74 [0.72,0.89]

TABLE VII

SIMULATION-BASED (SYNTHETIC) PAIRED COMPARISONS WITH Φ, EXACT MCNEMAR TESTS ON DISCORDANT PAIRS b/c (b: ONLY THE OTHER CONFIGURATION; c: ONLY Φ). FC: EPISODE ENDS IN A FALSE COMPLETION (100 SCENARIOS). DEC.: FIRST DECISION CORRECT (74 CLAIMS). SUCC.: TASK ACHIEVED AND REPORTED (100 SCENARIOS). MC: MEAN FCR [2.5–97.5TH PERCENTILES]; MEAN DECISION ACCURACY, OVER 1000 REGENERATED SCENARIO SETS WITH RATES DRAWN FROM THEIR BETA POSTERIORS. SYNTHETIC p-VALUES AND INTERVALS ARE CONDITIONAL ON THE SIMULATION ASSUMPTIONS; THEY ARE NOT EMPIRICAL TESTS OF THE AGENT.

Config.	FC b/c	p	Dec. b/c	p	Succ. b/c	p	MC FCR; dec. acc.
None	9/0	0.00391	3/9	0.146	3/1	0.625	.36 [.25,.48]; .64
Basic	5/0	0.0625	0/5	0.0625	0/1	1	.33 [.21,.45]; .66
Φ	ref.		ref.		ref.		.25 [.14,.37]; .74
Enhanced‡	0/11	¡0.001	11/2	0.0225	4/2	0.688	.14 [.04,.24]; .79

TABLE VIII

SIMULATION-BASED (SYNTHETIC) STUDY C: ONE COMPONENT REMOVED AT A TIME, ON THE SAME 100 SCENARIOS (POLICY LEVEL). EACH ROW IS A SIMULATED CONFIGURATION, NOT A RUN OF THE IMPLEMENTATION; EACH CORRESPONDS TO A SWITCH IN THE CODE (`VERIFICATION_MODE`, `ENABLE_MEMORY`, `ENABLE_RECOVERY`). §AS IN THE CODE, DISABLING THE RECOVERY ENGINE LEAVES THE THREE INLINE RETRIES OF THE VERIFICATION NODE IN PLACE. REMOVAL OF STORED EVIDENCE IS NOT SIMULATED: Φ READS THE LIVE PAGE, SO STORED EVIDENCE AFFECTS THE AUDIT TRAIL, NOT DECISIONS (CODE BEHAVIOUR). SUCC.: PAIRED TASK-SUCCESS DISCORDANCE b/c WITH Φ AND EXACT MCNEMAR p. MC: MEANS OVER 1000 REGENERATED SETS. SYNTHETIC p-VALUES AND INTERVALS ARE CONDITIONAL ON THE SIMULATION ASSUMPTIONS; THEY ARE NOT EMPIRICAL TESTS OF THE AGENT.

Configuration	TP (of 100)	FCR	FRR	F1	Succ. b/c; p	MC TSR	MC FCR
Full (Φ)	44 [.35,.54]	.302	.120	.779	ref.	0.47	0.25
– Φ	46 [.37,.56]	.378	.061	.748	3/1; 0.625	0.46	0.36
– memory	44 [.35,.54]	.312	.120	.772	1/1; 1	0.46	0.25
– recovery engine§	35 [.26,.45]	.314	.146	.761	0/9; 0.00391	0.42	0.25

The direction is fixed by construction: Φ can only withhold a claim that would otherwise be accepted, so no scenario can end in a false completion with Φ alone, one discordant cell is always 0, and only the size of the reduction is informative. Across 1,000 regenerated scenario sets, with the derived rates drawn from their posteriors, the reduction averages 0.11 false completions per scenario (2.5–97.5th percentiles 0.05–0.18), and the paired test gives $p < 0.05$ in 94% of the sets. Under the stated assumptions the simulation is consistent with H1; it is not empirical confirmation.

H2 (decision reliability). On the 74 claims, Φ decides correctly in 52 (accuracy 0.70, Wilson 0.59–0.79) and accepting every claim in 46 (0.62, 0.51–0.72): Φ is correct alone in 9 claims, all rejections of premature claims, and wrong alone in 3, all false rejections (exact McNemar $p = 0.15$). The difference has the hypothesized direction but is not significant in this set, and across regenerated sets Φ is better at $p < 0.05$ in only 42% (mean accuracy 0.74 against 0.64), so H2 is not supported for Φ at this sample size. The proposed enhanced validation reaches 61 of 74 (0.82; against Φ 11 and 2, $p = 0.02$), a difference that follows from its assumed detection rates. At the policy level, Φ completes 44 tasks against 46 without verification (discordant pairs 1 and 3, $p = 0.63$): false rejections cost a few successes, and continuation after a correct rejection recovers about as many.

B. Ablation (Study C)

Table VIII removes one component at a time. Removing Φ reproduces the no-verification row. Disabling the recovery engine, with the inline retries still in place, lowers TSR from 0.44 to 0.35 (9 and 0 discordant scenarios, $p = 0.004$) and, across replications, from 0.47 to 0.42; the size of this effect is set by the recovery probabilities of Study E, which rest on assumed per-attempt success rates. Removing memory changes one outcome in each direction and nothing on average. Stored evidence is not ablated: Φ reads the live page, so stored evidence serves the audit trail and does not change decisions.

C. Memory and Retrieval (Study D; H3)

On the synthetic memory corpus the real store returned a relevant record in the top three for each of the 9 queries that had one (recall@3 1.00, MRR 0.93 over those 9; ACTUAL). It has no relevance threshold, however: a query with no relevant memory still returned three records, and for 3 of 10 queries the top record was irrelevant or a *failed* earlier attempt (Table IX). Relevance is labelled by task group and is loose (a search for one place counts as relevant to a search for another), so recall is optimistic, and precision@3 is bounded by the number of relevant records per query. With that measured misleading rate, the simulated repeated workflows succeed 19 times of 44 with and without memory, so H3 is not supported. With 44

TABLE IX

STUDY D, MEMORY AND RETRIEVAL. UPPER BLOCK ACTUAL (REAL STORE, SYNTHETIC RECORDS); LOWER BLOCK SIMULATION-BASED (SYNTHETIC). P(MISLEADING): PROBABILITY THAT THE RETRIEVED CONTEXT MISLEADS THE PLANNER; FOR THE FILTER IT IS AN ASSUMED INPUT, SO ANY EFFECT OF THE FILTER FOLLOWS FROM THAT ASSUMPTION. WITH 44 REPEATED WORKFLOWS THE STUDY CANNOT DETECT AN EFFECT OF THE ASSUMED SIZE (GAIN 0.08, LOSS 0.10). ‡NOT IMPLEMENTED.

ACTUAL retrieval, real store, 22 synthetic records; 9 queries with a relevant record			
Precision@3 / Recall@3 / MRR	0.52	1.00	0.93
Top-1 irrelevant or failed (all 10 queries)	3/10		
SYNTHETIC repeated workflows (n=44)			
Configuration	TSR	FCR	P(misleading)
Memory disabled	19/44 [0.30,0.58]	.387	–
Memory enabled	19/44 [0.30,0.58]	.367	0.30 (meas.)
Relevance-filtered‡	20/44 [0.32,0.60]	.355	0.10 (assumed)

TABLE X

SIMULATION-BASED (SYNTHETIC) STUDY E: 120 FAILURE SCENARIOS, 20 PER MODE; 27 DRAWN AS PERMANENT (25% EXPECTED), NONE OF THEM A MISSING CONFIRMATION. ENTRY PATHS AND STRATEGY SEQUENCES FOLLOW THE CODE: ACTION ERRORS ARE RETRIED THREE TIMES INLINE BEFORE THE ENGINE IS CALLED; NAVIGATION FAILURES AND REQUESTS FOR HELP GO TO THE ENGINE DIRECTLY; A MISSING CONFIRMATION RAISES NO ERROR AND IS HANDLED ONLY BY REJECTION AND CONTINUATION. PER-ATTEMPT SUCCESS RATES ARE ASSUMED (PROPOSED FOR THE GUIDED VARIANT), SO DIFFERENCES BETWEEN ROWS FOLLOW FROM THEM. DETECTED: FAILURE NOTICED; FC: UNDETECTED MISSING CONFIRMATIONS REPORTED COMPLETE; UNNEC.: ATTEMPTS ON PERMANENT FAILURES; ATT.: MEAN ATTEMPTS. ‡NOT IMPLEMENTED.

Recovery	Detected	Recovered	FC	Unnec.	Att.
Inline retries only	111/120 [0.86,0.96]	29/93 [0.23,0.41]	9	51	1.28
Retry + engine (impl.)	111/120 [0.86,0.96]	67/93 [0.62,0.80]	9	162	3.12
Evidence-guided‡	118/120 [0.94,1.00]	87/93 [0.87,0.97]	2	162	3.17
Recovered / recoverable by mode (inline only; retry + engine; guided)					
nav. 0; 14; 16 / 16, time-out 9; 12; 13 / 14, missing el. 9; 13; 14 / 15					
invalid sub. 9; 13; 14 / 14, unexp. state 0; 13; 13 / 14, missing conf. 2; 2; 17 / 20					

workflows and an assumed gain of 0.08 the study could not have detected an effect of that size, so it is uninformative about H3 rather than evidence against it. The proposed relevance filter (20 of 44) changes the outcome only through its assumed misleading rate.

D. Recovery (Study E; H4)

The real recovery engine classifies a navigation error containing `net::ERR_` as transient, treats an invalid submission as a verification failure and assigns it re-planning, and does not recognize a missing confirmation, which it classifies as unknown (ACTUAL). In the code, action errors are first retried three times inline and only then passed to the engine, navigation failures and requests for help go to the engine directly, and a missing confirmation raises no error at all. Simulated along these paths (Table X), inline retries alone recover 29 of 93 recoverable failures and none of the navigation failures or unexpected states; with the engine, 67 of 93. Neither recovers missing confirmations (2 of 20, both by continuation), and 9 of them end as false completions. The proposed evidence-guided recovery, which checks an explicit confirmation postcondition, recovers 87 and leaves 2 false completions (20 against 0 discordant scenarios against retry); this is a consequence of its assumed success rates, not evidence for H4. Both engine variants spend 162 attempts on permanent failures.

TABLE XI

STUDY F, VERIFICATION OVERHEAD. UPPER BLOCK ACTUAL; P95 IS THE ORDER STATISTIC OF RANK $\mathrm{round}(0.95(n-1))+1$. *83 DECISIONS ON THE CONSTRUCTED GATE STATES, EACH THE MEDIAN OF 20 REPETITIONS (SECTION VIII-C). LOWER BLOCK SIMULATION-BASED (SYNTHETIC), BUILT FROM THE ACTUAL COMPONENT SAMPLES AND AN ASSUMED INFERENCE TIME, WITH RE-PLANNING AFTER REJECTIONS AND RECOVERY ATTEMPTS INCLUDED; THE TWO BLOCKS ARE NOT COMBINED INTO ONE CLAIM. INFERENCE TAKES 80% OF SYNTHETIC EPISODE TIME UNDER Φ, EVIDENCE SCREENSHOTS 4.1%. ‡NOT IMPLEMENTED.

ACTUAL component latency (ms), local sites, 30 repetitions		
Operation	median	p95
navigate	17.1	24.9
click + load	65.2	106.2
fill	12.5	17.4
DOM extraction	76.4	105.6
screenshot (evidence)	61.3	93.6
Φ on a site page	5.2	6.9
Φ on gate states*	26.8	30.3
SYNTHETIC episode time (s), inference median 3.0 s assumed		
Configuration	median	mean	verif. share
No verification	18.1	19.7	0.00%
Evidence-gated (Φ)	18.9	20.2	0.11%
LLM self-verification	21.2	22.4	12.21%
Φ + model check‡	22.6	24.0	14.13%

E. Verification Overhead (Study F; H5)

Φ takes a median of 5.2 ms on an ordinary page of the self-hosted sites and 26.8 ms on the constructed gate states; a screenshot 61 ms and DOM extraction 76 ms (ACTUAL, Table XI). In the simulation, with an assumed median of 3 s per model call on a CPU, inference takes 80% of an episode and the checks of Φ 0.1%. The mean episode is nevertheless 2.6% longer with Φ (20.2 against 19.7 s), because a rejected claim costs further planning steps; a model self-check adds 14% (22.4 s). At 1.5 s or 6 s per call the median episode under Φ takes 10.0 s or 36.8 s. H5 is consistent with the simulation: the rule itself is cheap, and its cost is in false rejections and the re-planning that rejections cause.

F. Robustness and Sensitivity (Study H)

The advantage of verification depends on how often the planner claims completion prematurely, which is unmeasured (Fig. 3): at $\pi = 0.3$ the FCR of Φ is 0.14 against 0.21 without verification, at $\pi = 0.95$ 0.32 against 0.45, while FRR stays near 0.14 against 0.08. Halving the derived detection rates of Φ raises its FCR from 0.25 to 0.30, about halfway to no verification (0.36); missing evidence raises FRR (0.09 to 0.22) because Φ fails closed on an unreadable page; more action failures lower TSR; and a higher rate of finishing after a rejection raises the TSR of Φ from 0.45 to 0.50 (Table XII).

X. DISCUSSION

a) What the evidence supports: A rule over observed state, evaluated inside the agent loop, withholds a significant

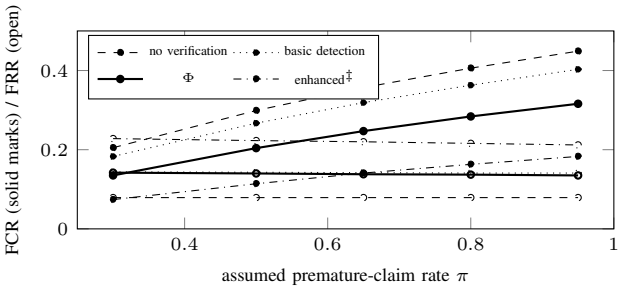


Fig. 3. SIMULATION-BASED (SYNTHETIC) Study H: mean FCR and FRR over 300 regenerated scenario sets per point as the assumed rate at which the planner claims completion of an unfinished task varies. Stricter verification lowers FCR at every rate and raises FRR; its FCR advantage grows with π. ‡Not implemented.

TABLE XII
SIMULATION-BASED (SYNTHETIC) STUDY H: ROBUSTNESS OF THE EVIDENCE-GATED CONFIGURATION (MEANS OVER 300 REGENERATED SETS PER ROW, DRAWN FROM A RANDOM STREAM SEPARATE FROM THE 1,000-SET MONTE CARLO OF TABLE VII, SO THE UNPERTURBED ROWS DIFFER SLIGHTLY FROM IT). WITHOUT VERIFICATION FCR STAYS AT 0.36 AT EVERY CONTINUATION RATE.

Varied factor (Φ)	FCR	FRR	TSR
Φ detection $\times 0.5$	0.30	0.14	0.46
Φ detection $\times 1.0$	0.25	0.14	0.47
Φ detection $\times 1.5$	0.19	0.13	0.49
missing evidence 0%	0.23	0.09	0.52
missing evidence 8%	0.25	0.14	0.47
missing evidence 20%	0.29	0.22	0.39
action failure rate 0.1	0.25	0.14	0.50
action failure rate 0.25	0.25	0.14	0.47
action failure rate 0.4	0.25	0.14	0.44
finish after rejection 0.1	0.26	0.14	0.45
finish after rejection 0.35	0.25	0.14	0.47
finish after rejection 0.6	0.24	0.13	0.50

share of premature completions on constructed browser states (41% of unsatisfied held-out states), with 1 of 20 satisfied held-out states rejected (upper 95% bound 0.24), at a cost of tens of milliseconds without model calls. Its defects showed up only when it was tested inside the live node, and its coverage on new phrasings is the binding limit. The evidence does not show that the agent completes tasks.

b) Rules and model judges: The held-out results argue against extending hand-written web rules indefinitely: each rule is keyed to the wording it was written for. Model-based judges [7], [26] cover broader goals but add inference cost and are models judging models [9]. The self-verification baseline in our harness will measure this alternative on the same tasks. We expect the two to be complementary—exact postconditions where the environment exposes structured state, a judge as a second opinion where it does not—with both inputs kept as evidence.

XI. SAFETY, PRIVACY AND AUTONOMY

a) Threat model: The user and the host are trusted; content the agent reads is not. We consider (i) instructions injected through web pages [31]; (ii) high-impact actions without consent, such as posting, sending, purchasing or deleting; (iii) exposure of credentials and of screenshots that show sensitive data; and (iv) disclosure of user data to parties other than the site the user asked the agent to use, in particular to a model provider. A compromised host, browser or model is out of scope.

TABLE XIII
PRIVACY AND SECURITY ASSESSMENT (STUDY G, ANALYTICAL). EVIDENCE: E EGRESS TEST, C CODE INSPECTION, T UNIT TEST; – NONE.

Surface	Control in the code	Evid.	Residual risk
Model inference	loopback endpoint enforced; override must be explicit	E, T	user override, or a loopback server that forwards requests (Ollama cloud models, a tunnel), sends prompts and screenshots off the machine; E used a stand-in server
Memory, task data	SQLite and Chroma on local disk; telemetry off	C, T	not encrypted; readable by any local process
Websites	none: the planner may follow links to any site	C	data the agent types reaches that site
Browser traffic	none in the agent; the test harness logs page-request URLs	E	Chromium background traffic, WebSockets and DNS not observed
Screenshots	stored in the local evidence folder	C	not redacted; may show account data
Retention	no deletion policy	C	evidence grows without bound
High-risk actions	approval before posting, sending, purchasing, deleting	C, T	model-assigned label; once per task
Injected text	17-pattern sanitizer on element text	C, T	denylist; titles and memory unfiltered

b) Privacy property: We call the system privacy-preserving in this sense: *by default the model client sends prompts and screenshots only to a model server on the loopback interface, and the code writes task records, memory and evidence only to local files.* The model client refuses a non-loopback endpoint unless the user sets an explicit override, Chroma telemetry is off, and in 70 harness episodes with a stand-in model server the agent's Python processes attempted no connection to a non-loopback address (Section VIII-D). The property does not cover: what the loopback model server does with a request (Ollama cloud models or a local tunnel forward it off the machine) and the server's own traffic; prompt context drawn from memory and retrieved documents, which goes to whatever endpoint is configured; data the user asks the agent to enter on a website, which reaches that website; the user-invoked search-engine fallback, which sends the query to that provider; model weights and Chroma's default embedding model, which are downloaded once; and unencrypted local storage. The egress test observed outbound `connect` calls in the harness's Python processes and the URLs of page requests reported by Playwright; it did not observe a real model server, the Playwright driver or Chromium processes (background traffic, WebSockets), or DNS lookups. Local file storage is established by code inspection, not by the egress test.

Table XIII summarizes each surface, the control in the code, the evidence for it and the residual risk.

c) Autonomy and what the implementation provides: The agent acts only in the browser and exposes no file, process or command action. Before navigation, a task whose parsed action is posting, e-mailing, purchasing, deleting or

transferring, or whose risk level is high or critical, waits for the user's approval; intent and risk come from the request alone. Bot checks stop the task for the user. A sanitizer replaces 17 instruction-override, delimiter, exfiltration and role-manipulation patterns in element text. The service listens on loopback, and each successful model completion is logged with destination and payload sizes (failed attempts and model-list calls are not, and the log can be disabled).

d) What it does not provide: The sanitizer is a fixed denylist, which systematic evaluations find insufficient [32]; titles, extracted text and memory are not filtered. The action label and risk level that trigger approval come from a small model and are decided once per task, so an injected instruction that steers the planner to submit a form mid-task is not gated. The API has no authentication, typed text is stored in clear, and screenshots are not redacted. We make no security claim. The browser hides common automation signals; we consider this acceptable only for a personal assistant acting at human pace on its user's requests, and deployments should disable it where a site's terms prohibit automated access.

XII. Limitations and Threats to Validity

a) Internal validity: One author wrote all gate states and labels and designed the fixes on the design states, so the design results are optimistic. Held-out 2 was frozen before testing, but its author knew the gate's rules; independent states and labels would give a less biased estimate. The live mode stands in for the planner with empty or generic reasoning, not model output.

b) External validity: All gate measurements use small, static pages served offline; real pages are larger and change, and latency will grow. No result involves a language model or a live website, so the agent is not shown to complete tasks. The end-to-end harness uses fictional, self-hosted sites, which avoids memorized answers and site drift but is simpler than real sites.

c) Statistical power: The sets are small: with 22 unsatisfied held-out 2 states the 95% interval of the withheld share spans 0.23–0.61, and the ablation cannot separate conjuncts that change one or two decisions.

d) Simulation validity: The simulated results follow from assumptions about planner behaviour, action failures and memory that no measurement supports; paired p-values describe the synthetic sample, not the agent. The measured inputs come from small constructed sets (54 held-out gate states, of which 27 unsatisfied and 27 satisfied; 10 memory queries; 6 failure messages), and the proposed components are simulated by assumed rates. The reported model (v2) was corrected after its first results and an audit were known, so it is not pre-registered; the corrections bring it closer to the code, but the choice of what to correct was made with the results in view. Its sample size (100 scenarios) is chosen, so synthetic significance is partly a matter of that choice. It can show which assumptions matter (Study H), not which outcome will occur.

e) Construct validity: FCR and FRR depend on the oracle. The browser predicate is keyed to request wording, so its coverage depends on how users phrase requests; when no keyword fires, it reduces to an error check.

f) System limitations: Planning with 1.5B–7B models is weak; vision-grounded clicks are not individually checked; two of the four recovery strategies have no distinct behavior; evidence files carry no integrity protection.

XIII. Future Work

The first step is to run the released end-to-end protocol with local models, which answers the open questions on task success, the gate's effect on the full agent, the self-verification baseline, memory and recovery. For the gate, we will derive per-step postconditions from the planner's expected outcomes instead of request wording, add a local vision-language judge as a second opinion, and have the held-out states labelled independently. Beyond the browser, the computer backend of Appendix A would extend the gate to desktop applications, files and processes, behind deterministic pre-action permission checks.

XIV. Conclusion

Agentic Pilot is a local browser agent built on the rule that an executed action is not a completed task: it may report completion only when a predicate over the observed page holds, and it keeps the inputs of that decision as evidence. The paper offers three kinds of evidence, which should be read separately. *Measured at component level:* inside the live loop the predicate withheld most premature completions on the states it was designed on and 41% on states frozen before testing, at a median of 27 ms and no model call, so its coverage of new phrasings, not its cost, is the limit; by default the model client connects only to a loopback endpoint, and in 70 episodes with a stand-in model server the agent's Python processes connected only to loopback addresses, which says nothing about where a real model server sends a request. *Simulated, conditional on stated assumptions:* the gate's benefit depends on how often the planner claims completion prematurely and is paid for in false rejections, and memory and recovery would need a relevance filter and an explicit confirmation check. *Not executed:* an end-to-end evaluation with a language model; the harness for it is released with an oracle validated on scripted trajectories, and its results will decide the questions the simulation can only frame.

Acknowledgment

An AI system (Anthropic's Claude) assisted with analyzing the source code, implementing the experiment harness, running the reported component experiments, searching and checking references, and drafting the text of this paper. The authors reviewed, verified and revised all content and take full responsibility for it.

TABLE XIV
COMPLETION POSTCONDITIONS BY ENVIRONMENT AND THEIR STATUS IN THE CODE. IMPL.: IMPLEMENTED AND USED BY THE AGENT; UNIT: IMPLEMENTED AND UNIT-TESTED, NOT CALLED BY THE AGENT; PROP.: PROPOSED DESIGN. h_0: HASH BEFORE THE ACTION; (pid, t_c): PROCESS ID AND CREATION TIME.

Env.	Example postcondition	Status
Browser	no error or sign-in page; results page for the quoted query; requested text present; answer mentions the goal; no pending step	Impl.
Browser	bot-check page detected ⇒ BLOCKED	Impl.
File	path exists and size $\geq n$	Unit
File	created / overwritten: sha256 = sha256(content); moved: source absent and sha256(dst) = h_0(src); trashed: path absent and trash entry present	Prop.
Process	launched: (pid, t_c) alive; terminated: (pid, t_c) gone within a timeout	Prop.
Window	foreground title matches; window with title exists	Prop.
Desktop	accessibility value of a control equals the input; control state contains *checked*	Prop.
UI		
Read-only	workspace manifest (paths, sizes, mtimes) unchanged	Prop.

TABLE XV
END-TO-END EVALUATION DESIGN (IMPLEMENTED IN `benchmarks/e2e`, NOT YET RUN WITH A MODEL). EACH CONFIGURATION CHANGES ONE FACTOR OF THE FULL SYSTEM; ALL USE THE SAME TASKS, MODEL, SAMPLING AND BUDGET.

Configuration	Change	Question
`full`	–	task success, FCR, FRR, cost
`no_verification`	every completion claim accepted	baseline (a); gain and overhead of Φ
`self_verification`	planner model judges its own completion	baseline (b)
`gate_without_`c	conjunct c removed	per-evidence ablation
`no_memory`	memory and RAG off	contribution of memory
`no_evidence`	no per-action execution records	traceability
`no_recovery`	failures not retried	contribution of recovery
`recovery_`s`_only`	strategy s pinned	per-strategy recovery
`single_model`, `vision_only`	routing off; no element list	routing, grounding

APPENDIX A
COMPUTER BACKEND (DESIGN, NOT IMPLEMENTED)

The repository contains a desktop executor that wraps PyAutoGUI and psutil, a text-file helper and a file-existence predicate. None is connected to the control loop: no node, action type or API route calls them, and the file helper does not confine paths. The rest of this appendix is a design.

a) Observation and grounding: The backend would observe the foreground window through its accessibility tree (Windows UI Automation, macOS Accessibility, AT-SPI) into a manifest shaped like the DOM manifest, falling back to a vision model on a window screenshot. File and process actions are grounded in typed state (paths, hashes, process identity).

b) Postconditions and permission tiers: Each mutating action carries exact postconditions (Table XIV) captured with before/after values. Because many local actions cannot be undone, a postcondition checked afterwards is no protection on its own, so an `action_gate` node between planning and execution would compute a tier from the concrete action: T0 (read-only) runs freely; T1 (reversible, with a recorded inverse) runs automatically only if the user enabled it; T2 (destructive) and T3 (network-capable programs, credentials) always need approval bound to a hash of the exact action. Paths outside allow-listed roots, inline interpreters and privilege elevation would be refused. A design in which untrusted content cannot change the control flow [37] is a stronger alternative.

APPENDIX B
END-TO-END TASK SET

The 40 tasks (`benchmarks/e2e/tasks.json`) are distributed as follows (development/test): extraction 2/8, form submission 2/7, multi-step 1/5, navigation 2/2, search 1/2, text entry 1/2, unsatisfiable 1/4; tiers 1–4 contain 3/6, 4/11, 3/12 and 0/1 tasks. The test split uses all five sites; the notes site appears only there. The harness (`run_e2e.py`) takes the split, the number of trials and a list of the configurations of Table XV; the analysis script (`analyze_e2e.py`) writes result tables with Wilson intervals, McNemar tests and bootstrap intervals.

APPENDIX C
SIMULATION PARAMETERS AND PROVENANCE

Table XVI lists every parameter of the simulation with its class. Assumed values were chosen to be neither favourable nor unfavourable to verification: the assumed premature-claim rate (0.65) and finish-after-rejection rate (0.35) are swept in Study H, and the rates that decide the gate's behaviour are derived from the held-out gate states, not from the design states on which the gate was tuned. The per-scenario data (`scenarios.csv`, `recovery_scenarios.csv`), the results, the evidence ledger with the source and class of every simulated number, and the independent validation report are in `benchmarks/simulation/out/heldout_v2/` (v1: `out/heldout_v1/`) and `artifacts/results/`. All are regenerated by `python -m benchmarks.simulation.simulate_v2` and `make_tables`; the independent validator of v1 contains absolute paths of the machine it ran on.

REFERENCES

[1] S. Zhou, F. F. Xu, H. Zhu, X. Zhou, R. Lo, A. Sridhar, X. Cheng, T. Ou, Y. Bisk, D. Fried, U. Alon, and G. Neubig, "WebArena: A realistic web environment for building autonomous agents," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[2] H. He, W. Yao, K. Ma, W. Yu, Y. Dai, H. Zhang, Z. Lan, and D. Yu, "WebVoyager: Building an end-to-end web agent with large multimodal models," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL) (Volume 1: Long Papers)*, 2024, pp. 6864–6890.

[3] T. Xie, D. Zhang, J. Chen, X. Li, S. Zhao, R. Cao, T. J. Hua, Z. Cheng, D. Shin, F. Lei, Y. Liu, Y. Xu *et al.*, "OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), Datasets and Benchmarks Track*, vol. 37, 2024.

TABLE XVI

SIMULATION PARAMETERS OF THE CORRECTED MODEL V2 AND THEIR PROVENANCE (BENCHMARKS/SIMULATION/PARAMS.JSON, SIMULATE_V2.PY). DERIVED RATES USE $(k + 0.5)/(n + 1)$ ON THE HELD-OUT GATE STATES; MONTE CARLO SETS DRAW THEM FROM BETA$(k + 0.5, n - k + 0.5)$.

Parameter	Value	Class
Scenarios per set; dev / held-out seed; MC sets	100; 101 / 202; 1000	design
Task categories (nav., search, form, data entry, extraction, multi-step)	0.15 / 0.15 / 0.20 / 0.15 / 0.20 / 0.15	ASSUMED
Outcome classes (successful, incomplete, incorrect, missing, contradictory, stale, ambiguous)	0.40 / 0.15 / 0.12 / 0.08 / 0.08 / 0.07 / 0.10	ASSUMED
P(task truly complete | class), same order	1.0 / 0.0 / 0.0 / 0.5 / 0.5 / 0.3 / 0.5	ASSUMED
Planner claims completion: if complete / if not (π)	0.92 / 0.65	ASSUMED
Φ withhold rate on unsatisfied states, by task category	navi. 0.36, sear. 0.50, form. 0.50, data. 0.38, extr. 0.07, mult. 0.63	DERIVED
Basic detection withhold rate, same	navi. 0.36, sear. 0.10, form. 0.07, data. 0.13, extr. 0.07, mult. 0.13	DERIVED
False rejection: Φ / basic	0/24 $\rightarrow$ 0.02 / 0.02 (satisfied states of the mapped categories; bot-check pages excluded)	DERIVED
Ambiguous-class detection factor	0.5	ASSUMED
Verifier rejects when evidence is missing (Φ and basic)	0.8	ASSUMED rate
Enhanced: extra catch / extra false rej. / contradiction rej.	0.5 / 0.05 / 0.7	PROPOSED
Finish after rejection (continuation is verified again)	0.35	ASSUMED
Action failure; recovered with / without engine	0.25; 0.65 / 0.27	ASSUMED; from Study E
Repeated workflows; memory gain / loss	0.4; 0.08 / 0.1	ASSUMED
Misleading retrieval: store / filtered	0.30 / 0.1	DERIVED / PROPOSED
Recovery: permanent share; max attempts	0.25; 6	ASSUMED
Recovery entry paths and strategy sequences	code (Table X)	ACTUAL
Recovery success per attempt	0.10–0.45 (re-plan), 0.05–0.35 (vision); guided 0.35–0.55	ASSUMED / PROPOSED
Missing-confirmation detection: retry / guided	0.63 / 0.9	DERIVED / PROPOSED
Steps per episode by category (same order as above)	2 / 3 / 5 / 3 / 3 / 6	ASSUMED
Inference time per call (median, log-sd)	3.0 s, 0.5	ASSUMED
Browser, DOM, screenshot, Φ latency	resampled from measurements	ACTUAL

[4] B. Zheng, B. Gou, J. Kil, H. Sun, and Y. Su, "GPT-4V(ision) is a generalist web agent, if grounded," in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 235, 2024, pp. 61 349–61 385.

[5] C. Zhang, L. Li, S. He, X. Zhang, B. Qiao, S. Qin, M. Ma, Y. Kang, Q. Lin, S. Rajmohan, D. Zhang, and Q. Zhang, "UFO: A UI-focused agent for Windows OS interaction," in *Proc. 2025 Conf. Nations of the Americas Chapter Assoc. Comput. Linguistics: Human Language Technologies (NAACL-HLT) (Volume 1: Long Papers)*, 2025, pp. 597–622.

[6] N. Lukas, A. Salem, R. Sim, S. Tople, L. Wutschitz, and S. Zanella-Béguelin, "Analyzing leakage of personally identifiable information in language models," in *Proc. IEEE Symp. Secur. Privacy (SP)*, 2023, pp. 346–363.

[7] J. Pan, Y. Zhang, N. Tomlin, Y. Zhou, S. Levine, and A. Suhr, "Autonomous evaluation and refinement of digital agents," in *Proc. Conf. Language Modeling (COLM)*, 2024.

[8] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[9] J. Huang, X. Chen, S. Mishra, H. S. Zheng, A. W. Yu, X. Song, and D. Zhou, "Large language models cannot self-correct reasoning yet," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[10] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 24 824–24 837.

[11] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, and T. Scialom, "Toolformer: Language models can teach themselves to use tools," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, 2023.

[12] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, 2023.

[13] G. Kim, P. Baldi, and S. McAleer, "Language models can solve computer tasks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, 2023.

[14] G. Wang, Y. Xie, Y. Jiang, A. Mandlekar, C. Xiao, Y. Zhu, L. Fan, and A. Anandkumar, "Voyager: An open-ended embodied agent with large language models," *Trans. Mach. Learn. Res. (TMLR)*, 2024.

[15] X. Deng, Y. Gu, B. Zheng, S. Chen, S. Stevens, B. Wang, H. Sun, and Y. Su, "Mind2Web: Towards a generalist agent for the web," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023.

[16] J. Yang, H. Zhang, F. Li, X. Zou, C. Li, and J. Gao, "Set-of-mark prompting unleashes extraordinary visual grounding in GPT-4V," arXiv:2310.11441, 2023.

[17] J. Y. Koh, R. Lo, L. Jang, V. Duvvur, M. C. Lim, P.-Y. Huang, G. Neubig, S. Zhou, R. Salakhutdinov, and D. Fried, "VisualWebArena: Evaluating multimodal agents on realistic visual web tasks," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL)*, 2024, pp. 881–905.

[18] A. Drouin, M. Gasse, M. Caccia, I. H. Laradji, M. Del Verme, T. Marty, D. Vazquez, N. Chapados, and A. Lacoste, "WorkArena: How capable are web agents at solving common knowledge work tasks?" in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 235, 2024, pp. 11 642–11 662.

[19] R. Bonatti, D. Zhao, F. Bonacci *et al.*, "Windows agent arena: Evaluating multi-modal OS agents at scale," arXiv:2409.08264, 2024.

[20] S. Agashe, J. Han, S. Gan, J. Yang, A. Li, and X. E. Wang, "Agent S: An open agentic framework that uses computers like a human," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2025.

[21] C. Zhang, Z. Yang, J. Liu, Y. Li, Y. Han, X. Chen, Z. Huang, B. Fu, and G. Yu, "AppAgent: Multimodal agents as smartphone users," in *Proc. 2025 CHI Conf. Human Factors Comput. Syst. (CHI)*, 2025, pp. 70:1–70:20.

[22] W. Hong, W. Wang, Q. Lv, J. Xu, W. Yu, J. Ji, Y. Wang, Z. Wang, Y. Dong, M. Ding, and J. Tang, "CogAgent: A visual language model for GUI agents," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 14 281–14 290.

[23] K. Cheng, Q. Sun, Y. Chu, F. Xu, L. YanTao, J. Zhang, and Z. Wu, "SeeClick: Harnessing GUI grounding for advanced visual GUI agents," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL) (Volume 1: Long Papers)*, 2024, pp. 9313–9332.

[24] Y. Qin, Y. Ye, J. Fang *et al.*, "UI-TARS: Pioneering automated GUI interaction with native agents," arXiv:2501.12326, 2025.

[25] Y. Lu, J. Yang, Y. Shen, and A. Awadallah, "OmniParser for pure vision based GUI agent," arXiv:2408.00203, 2024.

[26] M. Sumyk and O. Kosovan, ""are we done yet?": A vision-based judge for autonomous task completion of computer use agents," arXiv:2511.20067, 2025.

[27] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[28] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. 2019 Conf. Empirical Methods Natural Lang. Process. and 9th Int. Joint Conf. Natural Lang. Process. (EMNLP-IJCNLP)*, 2019, pp. 3982–3992.

[29] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proc. 36th Annu. ACM Symp. User Interface Softw. Technol. (UIST)*, 2023.

[30] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," arXiv:2310.08560, 2023.

[31] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proc. 16th ACM Workshop Artif. Intell. Secur. (AISec)*, 2023.

[32] Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong, "Formalizing and benchmarking prompt injection attacks and defenses," in *Proc. 33rd USENIX Secur. Symp. (USENIX Security)*, 2024, pp. 1831–1847.

[33] Q. Zhan, Z. Liang, Z. Ying, and D. Kang, "InjecAgent: Benchmarking indirect prompt injections in tool-integrated large language model agents," in *Findings Assoc. Comput. Linguistics: ACL 2024*, 2024, pp. 10 471–10 506.

[34] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, and F. Tramèr, "AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), Datasets and Benchmarks Track*, vol. 37, 2024.

[35] Y. Ruan, H. Dong, A. Wang, S. Pitis, Y. Zhou, J. Ba, Y. Dubois, C. J. Maddison, and T. Hashimoto, "Identifying the risks of LM agents with an LM-emulated sandbox," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.

[36] T. Kuntz, A. Duzan, H. Zhao, F. Croce, Z. Kolter, N. Flammarion, and M. Andriushchenko, "OS-Harm: A benchmark for measuring safety of computer use agents," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS) Datasets and Benchmarks Track*, vol. 38, 2025.

[37] E. Debenedetti, I. Shumailov, T. Fan, J. Hayes, N. Carlini, D. Fabian, C. Kern, C. Shi, A. Terzis, and F. Tramèr, "Defeating prompt injections by design," arXiv:2503.18813, 2025.

[38] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu *et al.*, "Qwen2.5 technical report," arXiv:2412.15115, 2024.

[39] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai *et al.*, "Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution," arXiv:2409.12191, 2024.

[40] J. Lin, J. Tang, H. Tang, S. Yang, W.-M. Chen, W.-C. Wang, G. Xiao, X. Dang, C. Gan, and S. Han, "AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration," in *Proc. Mach. Learn. Syst. (MLSys)*, vol. 6, 2024.

[41] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J. Wen, "A survey on large language model based autonomous agents," *Front. Comput. Sci.*, vol. 18, no. 6, p. 186345, 2024.

[42] Z. Xi, W. Chen, X. Guo, W. He, Y. Ding, B. Hong *et al.*, "The rise and potential of large language model based agents: A survey," *Sci. China Inf. Sci.*, vol. 68, p. 121101, 2025.

[43] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

[44] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.

[45] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[46] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[47] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.